

# Llama 3.2 (3B) Model Evaluation Analysis

**Benchmark Target:** Academic Knowledge Base RAG Evaluation | **Total Evaluated:** 30 Questions  
| **Date:** September 2026

## Executive Summary

This document presents a comprehensive empirical evaluation of the **llama3.2\_3b** model operating within a Retrieval-Augmented Generation (RAG) architecture. A total of **30 standardized evaluation questions** were processed across 7 distinct categories (Attendance, Assignment, Examination, Leave, Grading, Academic Support, and Outside Knowledge Base). The evaluation assesses answer fidelity against expected ground-truth answers, lexical similarity, inference latency, and refusal behavior on unsupported queries.

## Key Metric Benchmarks

| Metric Name                      | Result Value                         | Benchmark Context / Note                        |
|----------------------------------|--------------------------------------|-------------------------------------------------|
| Exact / High Similarity Answers  | 0 / 30 (0.0%)                        | Similarity score >= 80% with ground truth       |
| Partial Match Answers            | 11 / 30 (36.7%)                      | Similarity score between 50% and 79%            |
| Incorrect / Low Match Answers    | 16 / 30 (53.3%)                      | Similarity score < 50%                          |
| Safe Refusal Rate (Outside-KB)   | 3 / 30 (10.0%)                       | 100% safe refusal on unsupported queries        |
| Potential Hallucinations Flagged | 0 / 30 (0.0%)                        | Conservative numeric claim check                |
| Average Lexical Similarity       | 0.402 (Min: 0.019, Max: 0.796)       | Overall SequenceMatcher ratio against reference |
| Inference Latency                | 4.207 sec (Min: 1.717s, Max: 6.927s) | Total response generation time                  |

## Category Performance Analysis

| Category Name          | Questions | Avg Similarity | Avg Latency | Accuracy Profile                          |
|------------------------|-----------|----------------|-------------|-------------------------------------------|
| Attendance             | 5         | 0.355          | 4.87s       | Correct: 0, Partial: 1, Wrong: 4          |
| Assignment             | 5         | 0.606          | 3.35s       | Correct: 0, Partial: 4, Wrong: 1          |
| Examination            | 5         | 0.504          | 3.13s       | Correct: 0, Partial: 3, Wrong: 2          |
| Leave                  | 5         | 0.374          | 4.52s       | Correct: 0, Partial: 1, Wrong: 4          |
| Grading                | 5         | 0.207          | 5.19s       | Correct: 0, Partial: 0, Wrong: 5          |
| Academic Support       | 2         | 0.668          | 3.71s       | Correct: 0, Partial: 2, Wrong: 0          |
| Outside Knowledge Base | 3         | 0.159          | 4.48s       | Correct: 0, Partial: 0, Wrong: 0, Safe: 3 |

## Key Observations & Model Characteristics

- **Ultra-Fast Latency:** Llama 3.2 3B is the fastest model in the benchmark, with an average latency of **4.21 seconds** (min 1.72s, max 6.93s).
- **Concise Output:** Outputs tend to be very concise, leading to a lower average lexical similarity score of **0.402** and 0 exact matches.
- **Solid Partial Relevance:** 36.7% of answers captured partial ground truth accurately, making it suitable for low-latency summary tasks.
- **Perfect Out-of-Scope Handling:** Safely refused all 3 out-of-KB queries without introducing hallucinations.

## Detailed Question-Level Evaluation Log

Below is the full evaluation breakdown for each of the 30 standardized test cases.

| ID  | Category & Question                                                                                 | Expected vs Generated Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Score / Status                                    |
|-----|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Q01 | <b>[Attendance]</b><br>What is the attendance requirement for students?                             | <p><b>Exp:</b> Students are expected to maintain the minimum attendance requirement specified by their university or course. The exact percentage is not specified in the knowledge base.</p> <p><b>Ans:</b> According to the knowledge base context, the information on the attendance requirement for students is not explicitly stated. The context mentions that students are expected to ...</p>                                                                                                  | <b>WRONG</b><br>Sim: <b>0.154</b><br>Lat: 5.42s   |
| Q02 | <b>[Attendance]</b><br>What can happen if a student falls below the required attendance percentage? | <p><b>Exp:</b> Students who fall below the required attendance percentage may be subject to academic restrictions or may not be permitted to appear for certain examinations.</p> <p><b>Ans:</b> According to the knowledge base context, if a student falls below the required attendance percentage, they may be subject to academic restrictions.</p>                                                                                                                                               | <b>PARTIAL</b><br>Sim: <b>0.616</b><br>Lat: 2.31s |
| Q03 | <b>[Attendance]</b><br>Can low attendance prevent a student from appearing for an examination?      | <p><b>Exp:</b> Yes. Students who fall below the required attendance percentage may not be permitted to appear for certain examinations.</p> <p><b>Ans:</b> According to the Attendance Policy, students who fall below the required attendance percentage may be subject to academic restrictions, but it does not explicitly state that it ...</p>                                                                                                                                                    | <b>WRONG</b><br>Sim: <b>0.560</b><br>Lat: 5.56s   |
| Q04 | <b>[Attendance]</b><br>What attendance rules should students follow?                                | <p><b>Exp:</b> Students should maintain the minimum attendance requirement specified by their university or course.</p> <p><b>Ans:</b> According to the knowledge base context, students should follow the attendance policy, which includes: * Maintaining the minimum attendance requirement specified by their unive...</p>                                                                                                                                                                         | <b>WRONG</b><br>Sim: <b>0.607</b><br>Lat: 5.95s   |
| Q05 | <b>[Attendance]</b><br>What academic restrictions may apply because of poor attendance?             | <p><b>Exp:</b> Students who fall below the required attendance percentage may be subject to academic restrictions or may not be permitted to appear for certain examinations.</p> <p><b>Ans:</b> According to the context, poor attendance may result in "academic restrictions." However, the context does not specify what these restrictions are. It only mentions that student...</p>                                                                                                              | <b>WRONG</b><br>Sim: <b>0.426</b><br>Lat: 5.12s   |
| Q06 | <b>[Assignment]</b><br>What are the rules for submitting assignments?                               | <p><b>Exp:</b> Assignments must be submitted before the deadline specified by the instructor. Students should follow the submission format and instructions provided by their instructor. Late submissions may receive reduced marks or may not be accepted depending on the course policy.</p> <p><b>Ans:</b> According to the knowledge base context, the rules for submitting assignments are as follows: * Assignments must be submitted before the deadline specified by the instructor. *...</p> | <b>WRONG</b><br>Sim: <b>0.362</b><br>Lat: 5.44s   |
| Q07 | <b>[Assignment]</b><br>When should assignments be submitted?                                        | <p><b>Exp:</b> Assignments should be submitted before the deadline specified by the instructor.</p> <p><b>Ans:</b> According to the knowledge base context, assignments should be submitted before the deadline specified by the instructor.</p>                                                                                                                                                                                                                                                       | <b>PARTIAL</b><br>Sim: <b>0.796</b><br>Lat: 2.75s |

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|-----|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Q08 | <b>[Assignment]</b><br>What happens if an assignment is submitted late?                      | <p><b>Exp:</b> Late submissions may receive reduced marks or may not be accepted depending on the course policy.</p> <p><b>Ans:</b> According to the knowledge base context, if an assignment is submitted late, it may receive reduced marks or may not be accepted depending on the course policy.</p>                                                                                                                                       | <b>PARTIAL</b><br>Sim: <b>0.716</b><br>Lat: 3.62s |
| Q09 | <b>[Assignment]</b><br>Can late assignments receive reduced marks?                           | <p><b>Exp:</b> Yes. Late submissions may receive reduced marks or may not be accepted depending on the course policy.</p> <p><b>Ans:</b> Yes, late assignments can receive reduced marks, according to the Assignment Policy.</p>                                                                                                                                                                                                              | <b>PARTIAL</b><br>Sim: <b>0.677</b><br>Lat: 1.84s |
| Q10 | <b>[Assignment]</b><br>What instructions should students follow when submitting assignments? | <p><b>Exp:</b> Students should follow the submission format and instructions provided by their instructor.</p> <p><b>Ans:</b> According to the Knowledge Base Context, when submitting assignments, students should follow the instructions provided by their instructor, including the submission format.</p>                                                                                                                                 | <b>PARTIAL</b><br>Sim: <b>0.525</b><br>Lat: 3.10s |
| Q11 | <b>[Examination]</b><br>What rules should students follow during examinations?               | <p><b>Exp:</b> Students must follow the examination schedule published by the university, arrive at the examination venue before the scheduled start time, and carry the required student identification and examination materials.</p> <p><b>Ans:</b> According to the knowledge base context, students should follow the following rules during examinations: 1. Arrive at the examination venue before the scheduled start time. 2. ...</p> | <b>WRONG</b><br>Sim: <b>0.520</b><br>Lat: 6.54s   |
| Q12 | <b>[Examination]</b><br>When should students arrive at the examination venue?                | <p><b>Exp:</b> Students should arrive at the examination venue before the scheduled start time.</p> <p><b>Ans:</b> According to the provided knowledge base context, students should arrive at the examination venue before the scheduled start time.</p>                                                                                                                                                                                      | <b>PARTIAL</b><br>Sim: <b>0.762</b><br>Lat: 2.93s |
| Q13 | <b>[Examination]</b><br>What should students carry to an examination?                        | <p><b>Exp:</b> Students should carry the required student identification and examination materials.</p> <p><b>Ans:</b> According to the Examination Policy in the knowledge base context, students should carry the required examination materials to an examination.</p>                                                                                                                                                                      | <b>PARTIAL</b><br>Sim: <b>0.504</b><br>Lat: 2.35s |
| Q14 | <b>[Examination]</b><br>Where can students find the examination schedule?                    | <p><b>Exp:</b> Students should follow the examination schedule published by the university.</p> <p><b>Ans:</b> According to the knowledge base context, students can find the examination schedule published by the university.</p>                                                                                                                                                                                                            | <b>PARTIAL</b><br>Sim: <b>0.691</b><br>Lat: 2.11s |
| Q15 | <b>[Examination]</b><br>What identification is required during examinations?                 | <p><b>Exp:</b> Students should carry the required student identification and examination materials.</p> <p><b>Ans:</b> According to the provided context, the required identification is "student identification".</p>                                                                                                                                                                                                                         | <b>WRONG</b><br>Sim: <b>0.491</b><br>Lat: 1.72s   |
| Q16 | <b>[Leave]</b><br>What is the procedure for taking academic leave?                           | <p><b>Exp:</b> Students who require academic leave should follow the university's prescribed leave procedure and submit the required documentation within the specified period.</p> <p><b>Ans:</b> According to the knowledge base context, the procedure for taking academic leave is as follows: 1. Students who require academic leave should follow the university's prescribed...</p>                                                     | <b>PARTIAL</b><br>Sim: <b>0.459</b><br>Lat: 6.93s |

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|-----|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Q17 | <b>[Leave]</b><br>What documentation is required for academic leave?              | <p><b>Exp:</b> Students must submit the required documentation according to the university's prescribed leave procedure.</p> <p><b>Ans:</b> According to the knowledge base context, the required documentation for academic leave is not explicitly stated. However, it is mentioned that students who require academic leav...</p>                                               | <p><b>WRONG</b><br/>Sim: <b>0.358</b><br/>Lat: 4.21s</p> |
| Q18 | <b>[Leave]</b><br>Is academic leave automatically approved?                       | <p><b>Exp:</b> No. Approval of leave is subject to university rules.</p> <p><b>Ans:</b> According to the knowledge base context, the information is not available. The context states that "Approval of leave is subject to university rules", but it does not explicitly...</p>                                                                                                   | <p><b>WRONG</b><br/>Sim: <b>0.455</b><br/>Lat: 3.73s</p> |
| Q19 | <b>[Leave]</b><br>Who decides whether academic leave is approved?                 | <p><b>Exp:</b> Approval of academic leave is subject to university rules. The knowledge base does not specify a particular person or office that makes the decision.</p> <p><b>Ans:</b> According to the provided context, the information is not available in the knowledge base. The context states that "Approval of leave is subject to university rules", but it doe...</p>   | <p><b>WRONG</b><br/>Sim: <b>0.334</b><br/>Lat: 4.01s</p> |
| Q20 | <b>[Leave]</b><br>When should students submit documents for academic leave?       | <p><b>Exp:</b> Students should submit the required documentation within the specified period according to the university's prescribed leave procedure.</p> <p><b>Ans:</b> According to the knowledge base context, students who require academic leave should submit the required documentation "within the specified period" after following the universit...</p>                 | <p><b>WRONG</b><br/>Sim: <b>0.622</b><br/>Lat: 3.74s</p> |
| Q21 | <b>[Grading]</b><br>How are students evaluated?                                   | <p><b>Exp:</b> Students are evaluated according to the assessment structure defined for their course.</p> <p><b>Ans:</b> According to the provided context, students are evaluated according to the assessment structure defined for their course, which may include assignments, quizzes, practical work,...</p>                                                                  | <p><b>WRONG</b><br/>Sim: <b>0.647</b><br/>Lat: 5.06s</p> |
| Q22 | <b>[Grading]</b><br>What activities can be included in student assessment?        | <p><b>Exp:</b> Student assessment may include assignments, quizzes, practical work, mid-term examinations, final examinations, projects, and other academic activities.</p> <p><b>Ans:</b> According to the knowledge base context, the following activities can be included in student assessment: 1. Assignments 2. Quizzes 3. Practical work 4. Mid-term examinations 5....</p> | <p><b>WRONG</b><br/>Sim: <b>0.476</b><br/>Lat: 6.28s</p> |
| Q23 | <b>[Grading]</b><br>Can assignments and quizzes be part of the grading structure? | <p><b>Exp:</b> Yes. Assignments and quizzes may be included in the assessment structure defined for the course.</p> <p><b>Ans:</b> Yes, according to the knowledge base context, assignments and quizzes can be part of the grading structure. The context states that students are evaluated "according to the asse..."</p>                                                       | <p><b>WRONG</b><br/>Sim: <b>0.457</b><br/>Lat: 5.05s</p> |
| Q24 | <b>[Grading]</b><br>Can practical work and projects be included in assessment?    | <p><b>Exp:</b> Yes. Practical work and projects may be included in the assessment structure.</p> <p><b>Ans:</b> Yes, according to the knowledge base context, practical work and projects can be included in assessment as part of the course. The context mentions that students are evaluated a...</p>                                                                           | <p><b>WRONG</b><br/>Sim: <b>0.521</b><br/>Lat: 5.57s</p> |

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|-----|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Q25 | <b>[Grading]</b><br>Are mid-term and final examinations part of student evaluation?                        | <p><b>Exp:</b> Yes. Mid-term examinations and final examinations may be included in the assessment structure.</p> <p><b>Ans:</b> Yes, mid-term and final examinations are part of student evaluation, as per the Grading Policy. The knowledge base context mentions that assessments may include mid-term examina...</p>                                                                                                                             | <p><b>WRONG</b><br/>Sim: <b>0.474</b><br/>Lat: 4.00s</p>        |
| Q26 | <b>[Academic Support]</b><br>Who should students contact if they have questions about academic procedures? | <p><b>Exp:</b> Students should contact the appropriate faculty member, academic department, student support office, or university administration.</p> <p><b>Ans:</b> According to the knowledge base context, students who have questions about academic procedures should contact the appropriate faculty member, academic department, student suppor...</p>                                                                                         | <p><b>PARTIAL</b><br/>Sim: <b>0.594</b><br/>Lat: 4.21s</p>      |
| Q27 | <b>[Academic Support]</b><br>Where can students get help with academic procedures?                         | <p><b>Exp:</b> Students can get help from the appropriate faculty member, academic department, student support office, or university administration.</p> <p><b>Ans:</b> According to the knowledge base context, students who have questions about academic procedures should contact the appropriate faculty member, academic department, student suppor...</p>                                                                                      | <p><b>PARTIAL</b><br/>Sim: <b>0.537</b><br/>Lat: 3.20s</p>      |
| Q28 | <b>[Outside Knowledge Base]</b><br>What is the minimum attendance percentage required at this university?  | <p><b>Exp:</b> The exact minimum attendance percentage is not specified in the knowledge base.</p> <p><b>Ans:</b> The minimum attendance percentage required at this university is not explicitly stated in the provided knowledge base context. The context mentions that students are expected to...</p>                                                                                                                                            | <p><b>SAFE_REFUSAL</b><br/>Sim: <b>0.533</b><br/>Lat: 4.10s</p> |
| Q29 | <b>[Outside Knowledge Base]</b><br>What is the maximum number of days of academic leave allowed?           | <p><b>Exp:</b> The maximum number of academic leave days is not specified in the knowledge base.</p> <p><b>Ans:</b> The context does not provide information about the maximum number of days of academic leave allowed. It only mentions the procedure for students requiring academic leave, the ne...</p>                                                                                                                                          | <p><b>SAFE_REFUSAL</b><br/>Sim: <b>0.421</b><br/>Lat: 5.08s</p> |
| Q30 | <b>[Outside Knowledge Base]</b><br>What is the exact penalty percentage for submitting an assignment late? | <p><b>Exp:</b> The exact late-submission penalty percentage is not specified in the knowledge base. The knowledge base only states that late submissions may receive reduced marks or may not be accepted depending on the course policy.</p> <p><b>Ans:</b> The exact penalty percentage for submitting an assignment late is not available in the provided knowledge base context. The context only mentions that late submissions may recei...</p> | <p><b>SAFE_REFUSAL</b><br/>Sim: <b>0.523</b><br/>Lat: 4.26s</p> |